

A Comparative Evaluation of AI-Powered Literature Discovery Tools Across Disciplines

Abstract

The rapid growth of scholarly publishing has made the discovery, screening, organization, and interpretation of relevant literature increasingly difficult. Conventional keyword-based search systems remain essential, but they can impose substantial cognitive and temporal burdens on researchers, particularly when terminology varies across disciplines or when important work is connected through indirect citation relationships. Artificial intelligence (AI)-powered literature discovery tools have emerged as a complementary approach. These systems employ semantic search, natural language processing, citation-network analysis, machine learning, large language models, and automated information extraction to assist researchers in identifying and evaluating scholarly publications. This paper comparatively evaluates major AI-powered literature discovery approaches and representative tools, including Semantic Scholar, Elicit, ResearchRabbit, Connected Papers, scite, and evidence-oriented research assistants such as Consensus. The comparison focuses on discovery mechanisms, corpus and metadata dependence, citation analysis, synthesis capabilities, transparency, reproducibility, and disciplinary suitability. The analysis finds that no single tool performs optimally across the entire research workflow or across all academic disciplines. Semantic and evidence-extraction systems are particularly useful for structured searches and large evidence bases, whereas citation-network tools are effective for exploratory mapping and interdisciplinary discovery. Citation-context systems provide a distinct contribution by helping researchers interpret how publications are used by later work. However, substantial challenges remain concerning corpus coverage, disciplinary and linguistic bias, opaque ranking algorithms, hallucinated or inaccurate summaries, reproducibility, and the inappropriate use of AI-generated outputs as substitutes for source verification. The paper argues that AI-powered discovery should be understood as an augmentation of established scholarly search practices rather than a replacement for disciplinary databases, expert judgment, and critical reading. Future research should prioritize benchmark datasets spanning multiple disciplines and languages, transparent evaluation protocols, provenance-aware AI systems, and reproducible human-in-the-loop workflows.

Keywords— artificial intelligence; literature discovery; scholarly search; scientific paper recommendation; semantic search; citation networks; systematic review; research synthesis.

I. Introduction

The volume of scholarly literature has increased to a scale at which comprehensive manual discovery is difficult even within relatively specialized research domains. Researchers must identify relevant studies, distinguish foundational work from peripheral publications, trace conceptual and methodological relationships, evaluate the credibility of claims, and determine whether apparently conflicting findings are genuinely inconsistent. Traditional academic databases and search engines address part of this problem through keyword search, controlled vocabularies, citation indexes, and disciplinary classifications. Nevertheless, conventional retrieval can perform poorly when researchers do not know the terminology of an unfamiliar field or when relevant publications use substantially different language.

AI-powered literature discovery tools attempt to address these limitations through alternative representations of scholarly information. Instead of relying exclusively on exact lexical matching, such systems may use semantic representations of titles and abstracts, citation graphs, author and venue information, natural-language queries, recommendation algorithms, or large language models (LLMs). Surveys of scientific paper recommendation systems show that modern approaches increasingly combine textual, bibliographic, network, and user-related signals rather than treating literature discovery as a simple keyword-retrieval problem [1].

The contemporary ecosystem includes several distinct categories of tools. Semantic Scholar represents an AI-enhanced scholarly search and recommendation environment. Elicit emphasizes question-driven discovery and structured information extraction. ResearchRabbit and Connected Papers emphasize graph-based exploration of relationships among publications. scite focuses on citation contexts and the characterization of how publications are cited. Consensus represents an evidence-oriented question-answering approach that attempts to connect natural-language questions with relevant scholarly studies. These tools overlap in functionality but embody substantially different assumptions about what “discovering literature” means.

This paper evaluates these approaches comparatively and across disciplines. Its central research question is:

How do AI-powered literature discovery tools differ in their effectiveness, strengths, and limitations across different academic disciplines and stages of the research workflow?

The paper makes four contributions. First, it develops a comparative framework for evaluating AI literature discovery systems. Second, it examines the technical approaches underlying major categories of tools. Third, it analyzes disciplinary variation, particularly differences between STEM, medicine, social sciences, and humanities. Finally, it identifies research challenges and future directions for trustworthy and reproducible AI-assisted scholarly discovery.

II. Background and Related Work

A. From Information Retrieval to Scholarly Recommendation

Classical information retrieval generally treats a user's query and a document collection as objects to be matched and ranked. Scholarly search adds further dimensions, including authorship, publication venues, citation links, publication dates, disciplinary categories, and document structure. Paper recommendation systems therefore evolved beyond keyword matching toward content-based, collaborative, graph-based, and hybrid approaches [1].

Content-based systems estimate similarity from textual information such as titles, abstracts, keywords, or full text. Citation-based approaches exploit bibliographic relationships. Bibliographic coupling connects papers that cite common references, whereas co-citation connects papers that are cited together. Hybrid systems combine multiple signals to improve recommendation quality.

A major difficulty is that relevance itself is multidimensional. A paper can be relevant because it studies the same topic, uses the same method, supports or contradicts a claim, belongs to the same intellectual lineage, or provides an influential theoretical foundation. Consequently, a tool optimized for one form of relevance may perform poorly for another.

B. Semantic Scholarly Search

Semantic search attempts to retrieve documents based on conceptual similarity rather than exact word overlap. This is particularly valuable when terminology differs between disciplines or changes over time. AI-enhanced scholarly search systems can rank papers according to learned semantic representations and supplement search results with features such as concise summaries, citation information, influential-reference indicators, and recommendations.

Semantic approaches are especially useful during exploratory research. A researcher entering an unfamiliar field may know a conceptual question without knowing the precise vocabulary used by domain specialists. However, semantic relevance does not guarantee methodological relevance, evidentiary quality, or comprehensiveness. A semantically similar paper may concern a different population, research design, theoretical tradition, or disciplinary context.

C. Citation-Network Exploration

Citation networks provide a different model of scholarly discovery. Rather than beginning with a query alone, graph-based systems can begin with one or more known “seed” papers and identify related literature through references, citations, shared citation patterns, and similarity relationships.

ResearchRabbit and Connected Papers exemplify this exploratory model. Their interfaces make relationships among publications visually accessible and can support backward and forward citation searching. This approach is particularly useful for identifying intellectual clusters and for discovering literature that keyword search may miss. The available literature on AI research-support workflows describes ResearchRabbit as a citation-based mapping tool and Connected Papers as a graph-based system for exploring related publications [2].

D. AI Assistance in Systematic and Structured Reviews

AI is increasingly used not only to locate papers but also to assist with search strategy development, title and abstract screening, data extraction, and evidence synthesis. Reviews of AI-assisted systematic-review workflows indicate that AI methods can potentially support several stages of evidence synthesis, while also emphasizing the importance of reporting AI use to preserve reproducibility [3].

This development changes the definition of literature discovery. Discovery is no longer merely the retrieval of documents; it can include identifying candidate evidence, comparing studies, extracting structured variables, tracing citations, and generating preliminary summaries.

III. Comparative Evaluation Framework

A meaningful comparison requires criteria broader than the number of papers retrieved. This paper evaluates tools according to seven dimensions:

1. **Discovery mechanism** — keyword search, semantic search, citation traversal, graph exploration, or question answering.
2. **Evidence representation** — whether results are presented as ranked documents, networks, evidence tables, citation contexts, or synthesized answers.
3. **Support for screening and extraction** — the ability to assist with comparing multiple studies systematically.

4. **Citation intelligence** — the extent to which the tool analyzes citation relationships or contexts.
5. **Transparency and provenance** — whether users can inspect the underlying papers and trace generated outputs to sources.
6. **Reproducibility** — whether searches, ranking decisions, extraction procedures, and AI-assisted outputs can be documented and repeated.
7. **Disciplinary suitability** — the extent to which corpus coverage and system design fit different fields.

The comparison is qualitative rather than a benchmark experiment. Therefore, it does not claim that one tool is universally superior.

Table I

Comparative Characteristics of Major AI-Powered Literature Discovery Approaches

Tool/Approach	Primary Discovery Method	Major Strength	Major Limitation	Strongest Research Use
Semantic Scholar	Semantic search and scholarly recommendation	Broad exploratory discovery	Corpus and disciplinary coverage may vary	General literature discovery
Elicit	Natural-language search and structured extraction	Comparative evidence tables	AI-extracted information requires verification	Structured and evidence-oriented reviews
ResearchRabbit	Citation and author-network exploration	Iterative discovery from seed papers	Depends on existing bibliographic relationships	Mapping a research area
Connected Papers	Similarity and citation-network visualization	Visual identification of related clusters	Not designed as a comprehensive database search	Intellectual lineage and field exploration
scite	Citation-context analysis	Interpretation of how work is cited	Citation characterization does not replace full-text evaluation	Claim and citation investigation
Consensus-type systems	Evidence-oriented natural-language querying	Rapid orientation to empirical evidence	Simplification and selection bias are possible	Initial evidence scoping

IV. Main Thematic and Technical Discussion

A. Semantic Search and Representation Learning

Semantic literature discovery attempts to solve the vocabulary problem inherent in traditional search. A query concerning “workplace flexibility,” for example, may be relevant to papers using terms such as telework, remote work, hybrid work, distributed work, or location-independent employment. Embedding-based systems can represent these expressions as conceptually related even when their words differ.

The advantage is particularly apparent in interdisciplinary research. Concepts frequently migrate between fields while acquiring different terminology. A computer scientist, psychologist, and organizational researcher may investigate related phenomena while using different conceptual vocabularies. Semantic search can therefore support bridge-building across disciplinary boundaries.

However, semantic systems can also obscure why a paper was ranked. A document may be recommended because its language is statistically similar rather than because it is methodologically appropriate. Researchers must therefore distinguish **algorithmic similarity** from **scholarly relevance**.

B. Citation Networks as Knowledge Structures

Citation networks capture explicit relationships generated by scholarly practice. A reference can indicate intellectual influence, methodological dependence, criticism, replication, background information, or rhetorical positioning. Citation counts alone are therefore incomplete measures of importance.

Graph-based tools provide a useful visual complement to ranked lists. They can reveal:

- clusters of closely related publications;
- foundational or highly connected papers;
- bridges between research communities;
- earlier work connected to a contemporary topic;
- later work that develops or cites a seed publication.

This is particularly valuable during the early stages of a literature review, when the researcher is attempting to understand the structure of a field.

The principal weakness is historical dependence. Citation networks are necessarily shaped by publication and citation practices. Newly published work may have few links. Important work outside the indexed corpus may be invisible. Fields with extensive monograph traditions may also be represented less effectively than fields dominated by journal articles and conference proceedings.

C. Structured Evidence Extraction

Tools such as Elicit represent a transition from retrieval toward semi-automated evidence organization. Instead of simply returning a list of papers, such systems can assist users in comparing publications according to variables such as population, method, sample size, intervention, outcome, or stated limitations.

This is potentially valuable for systematic and scoping reviews. Yet automated extraction introduces a significant epistemic risk: a generated table can appear more authoritative than the underlying

extraction process warrants. A model may misunderstand a table, confuse intervention and control groups, infer information not explicitly reported, or extract an answer from the wrong section of a paper.

Accordingly, AI-generated extraction should be treated as **candidate extraction** rather than final evidence. Human verification remains necessary, especially when extracted values affect inclusion decisions, quantitative synthesis, or substantive conclusions. Reviews of AI-supported systematic-review processes similarly emphasize both efficiency gains and the need for transparent reporting and methodological oversight [3].

D. Citation Context and Claim Evaluation

A citation is not inherently positive. A paper may be cited because it is supported, challenged, corrected, replicated, criticized, or merely mentioned. Citation-context systems attempt to move beyond raw citation counts by examining the surrounding textual context.

This capability is particularly important for researchers evaluating influential or controversial findings. A highly cited paper may be influential because it established a method, because it produced a widely replicated result, or because later research repeatedly criticized it. Citation-context analysis can therefore improve scholarly interpretation.

Nevertheless, automated classification of citation intent is imperfect. Context can be ambiguous, ironic, field-specific, or dependent on information elsewhere in the article. Citation labels should therefore support critical reading rather than replace it.

E. Natural-Language Evidence Questioning

Question-answering systems allow researchers to formulate questions in ordinary language rather than constructing complex Boolean queries. This can lower barriers to initial exploration and may be especially useful for students or researchers entering a new domain.

However, natural-language interaction introduces two problems. First, the system may transform a complex research question into an apparently simple answer. Second, users may focus on the generated synthesis rather than inspecting the diversity and limitations of the underlying literature.

A well-designed scholarly AI interface should therefore expose provenance: users should be able to inspect which studies support each statement and determine whether the evidence base is representative.

V. Comparison Across Academic Disciplines

A. Natural Sciences and Engineering

AI-powered discovery tools are generally well suited to fields characterized by large volumes of journal articles, conference papers, structured metadata, and dense citation networks. Computer science is particularly compatible with such systems because conference proceedings, preprints, citations, and rapidly evolving terminology are extensively represented in machine-readable scholarly infrastructures.

Semantic search can help researchers navigate fast-moving terminology, while graph tools can reveal methodological and conceptual clusters. The principal challenge is recency. New publications, preprints, and rapidly changing research fronts may not yet have developed meaningful citation networks.

B. Medicine and Health Sciences

Medical and health research presents both opportunities and risks. Large bibliographic databases and relatively standardized article structures support machine processing. Questions concerning interventions, outcomes, populations, and study designs are potentially well suited to structured extraction.

At the same time, errors can have serious consequences. A literature-discovery system may retrieve observational studies alongside randomized trials or may summarize an association as though it were causal. Systematic-review guidance and domain-specific databases therefore remain essential.

AI can improve efficiency in screening and preliminary organization, but it should not independently determine evidence quality or clinical recommendations. Recent reviews of AI in systematic-review workflows emphasize its potential role across search, screening, extraction, and synthesis while underscoring the importance of transparent methods [3].

C. Social Sciences

The social sciences benefit from semantic search because concepts are often expressed through diverse theoretical traditions. A single topic may involve different vocabularies in psychology, sociology, economics, education, political science, and management.

Citation-network tools can reveal interdisciplinary connections, but disciplinary boundaries complicate interpretation. A paper may be central within one theoretical tradition yet nearly invisible in another. Moreover, books, reports, surveys, policy documents, and other non-journal materials may be important evidence. Corpus completeness is therefore a major concern.

D. Humanities

Humanities research exposes some of the strongest limitations of current AI literature discovery systems. Scholarship may depend heavily on monographs, book chapters, critical editions, archival materials, non-English publications, and interpretive traditions that are less completely represented in machine-readable bibliographic datasets.

Semantic similarity may also be difficult to evaluate in fields where meaning is historically and culturally situated. A tool may identify texts discussing similar concepts while overlooking works connected through historical influence, genre, language, or interpretive methodology.

Consequently, AI-powered discovery is likely to be more supplementary in the humanities than in highly indexed STEM disciplines. Specialized library catalogs, archival finding aids, bibliographies, and discipline-specific databases remain indispensable.

VI. Current Approaches and Developments

The current development of AI literature discovery can be characterized by five trends.

A. Integration of Retrieval and Synthesis

Earlier systems primarily recommended papers. Contemporary systems increasingly connect retrieval with summarization, comparison, extraction, and question answering. This reduces workflow fragmentation but also increases the risk that users will accept generated interpretations without checking the original studies.

B. Hybrid Retrieval Architectures

Modern systems increasingly combine lexical retrieval, semantic representations, citation graphs, metadata, and generative models. Hybrid architectures are preferable to single-signal systems because scholarly relevance is multifaceted.

C. Evidence-Grounded Generation

A central development is the attempt to ground generated responses in identifiable scholarly sources. This is an important improvement over unconstrained generative systems, but citation alone is insufficient. The cited source must actually support the generated claim, and users must be able to inspect the relevant context.

D. Human-in-the-Loop Research Workflows

The most defensible model is not fully automated literature review but **human-AI collaboration**. AI can reduce repetitive work, prioritize candidate papers, organize evidence, and suggest connections. Researchers remain responsible for methodological decisions, interpretation, and final verification.

E. Workflow Complementarity

The available tools increasingly appear complementary rather than interchangeable. A researcher may begin with broad semantic discovery, use graph exploration to map the field, employ structured extraction to compare studies, and use citation-context analysis to investigate contested claims. The research-support literature similarly describes these tools as supporting different stages rather than constituting a single universal replacement for conventional databases [2], [3].

VII. Research Challenges and Limitations

A. Corpus Coverage

A system cannot retrieve literature that is absent from its underlying corpus. Coverage varies by discipline, publication type, language, geography, publisher, and historical period. This limitation is particularly serious for interdisciplinary and humanities research.

B. Algorithmic Opacity

Users often cannot determine why a particular paper was recommended or why another paper was omitted. Proprietary ranking and generative pipelines make independent evaluation difficult.

C. Hallucination and Extraction Error

Generative systems can produce inaccurate summaries or incorrectly extracted information. Even when a cited paper genuinely exists, the system may misrepresent its findings. Original sources must therefore remain the authoritative objects of scholarly interpretation.

D. Reproducibility

AI-assisted searches may change as models, corpora, ranking algorithms, and interfaces evolve. A query performed today may produce different results later. Researchers should document search dates, exact prompts or queries, tool versions where available, filtering decisions, and verification procedures.

E. Disciplinary and Linguistic Bias

English-language and STEM-oriented corpora can systematically disadvantage scholarship published in other languages or formats. A system's apparent comprehensiveness may therefore conceal structural gaps.

F. Evaluation Problems

Many existing evaluations focus on recommendation accuracy while neglecting broader research outcomes. A useful tool should be assessed not only by whether it retrieves relevant papers but also by whether it improves recall, supports transparent workflows, reduces bias, and helps researchers make better decisions.

VIII. Research Gaps and Future Directions

Several gaps require sustained scholarly attention.

First, there is a need for **cross-disciplinary benchmark datasets**. Evaluations should include STEM, medicine, social sciences, humanities, multilingual scholarship, books, conference proceedings, and nontraditional research outputs.

Second, future systems should provide **provenance-aware outputs**. Every generated claim should ideally be traceable to a precise source location, enabling the user to inspect the relevant passage rather than merely seeing a paper-level citation.

Third, reproducibility should become a design requirement. Tools should allow users to export search strategies, retrieved sets, ranking information, timestamps, and AI-generated extraction records.

Fourth, systems should better represent **negative, contradictory, and uncertain evidence**. Literature discovery should not merely identify popular or highly cited papers; it should help researchers recognize disagreement, failed replication, methodological controversy, and underrepresented perspectives.

Fifth, future work should investigate **fairness in scholarly representation**. This includes language diversity, geographic representation, publication formats, and disciplines with weaker digital infrastructure.

Finally, researchers need controlled comparative studies involving real scholarly tasks. Future evaluations should compare AI-assisted and conventional workflows using measures such as recall, precision, time expenditure, error rates, reproducibility, cognitive workload, and the quality of final research syntheses.

IX. Conclusion

AI-powered literature discovery tools represent a significant development in scholarly information seeking. Semantic search can overcome vocabulary barriers; citation-network systems can reveal the structure of intellectual communities; structured extraction tools can accelerate evidence organization; citation-context analysis can improve interpretation of scholarly influence; and evidence-oriented question-answering interfaces can support rapid orientation to a research problem.

However, the comparative analysis demonstrates that no single tool is optimal across disciplines or research tasks. Tools that perform strongly in biomedical research or computer science may be less suitable for humanities scholarship. Systems designed for exploratory discovery may be inappropriate for exhaustive systematic searching, while tools capable of generating concise evidence summaries may obscure uncertainty or corpus limitations.

The most productive model is therefore an integrated workflow in which AI augments rather than replaces conventional scholarly practice. Researchers should combine AI-powered discovery with discipline-specific databases, citation searching, transparent documentation, expert judgment, and direct examination of original sources.

The future value of AI literature discovery will depend not simply on more powerful models, but on systems that are more transparent, reproducible, inclusive, provenance-aware, and aligned with the methodological standards of different academic disciplines.

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Note on references: The tools discussed in this paper evolve rapidly, and product features and underlying corpora can change over time. The paper therefore treats tool-specific functionality as an evolving implementation layer and bases its broader technical and methodological conclusions primarily on the scholarly literature concerning paper recommendation, semantic scientific representations, citation analysis, and AI-assisted systematic review workflows. The most directly verified recent scholarly sources used for the AI-tool discussion include the paper-recommendation survey and the PubMed Central articles on Elicit and AI-assisted systematic reviews.